Honors Differential Equations

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Supplimental 1

I find it much neater to write things with inner products and norms, rather than messy integrals. Especially when the integrals are in a fraction, because then they get squished down and look really ugly and I kinda hate it.

Note that $\lambda_n = 0$ if and only if $n = 0$. Also note that since $P_n \in L^2[-1, 1]$ for all $n \in \mathbb{N}$, where we have defined L^2 over $\mathbb{R}$, we have the inner product

$$\langle P_n, P_m \rangle = \int_{-1}^{1} P_n(t) P_m(t) \, \mathrm{d}t.$$

Let $n \neq m$ be natural numbers. By logic that has been explored previously, for the case of $\lambda_n, \lambda_m \neq 0$ it suffices to show $\lambda_n \langle P_n, P_m \rangle = \lambda_m \langle P_n, P_m \rangle$. The case of $\lambda_n = 0$ must be explored somewhat since we multiply through by λ_n at the beginning of the proof. Since $\lambda_n \neq \lambda_m$, at most one of these λ values can be 0, so without loss of generality let $\lambda_n \neq 0$. If we show that $\lambda_n \langle P_n, P_m \rangle = \lambda_m \langle P_n, P_m \rangle$ for the case of $\lambda_m = 0$, it follows that $\lambda_n \langle P_n, P_m \rangle = 0$, and thus $\langle P_n, P_m \rangle = 0$. Therefore, it suffices to show this is true for the general case.

Note that

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(\left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} \right) = \left(1 - t^2\right) \frac{\mathrm{d}^2 P_n}{\mathrm{d}t^2} - 2t \frac{\mathrm{d}P_n}{\mathrm{d}t}.$$

It follows that

$$\lambda_n P_n(t) = -\frac{\mathrm{d}}{\mathrm{d}t} \left(\left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} \right).$$

Now consider the following.

$$\begin{aligned} \lambda_n \langle P_n, P_m \rangle &= \int_{-1}^{1} P_n(t) P_m(t) \, \mathrm{d}t \\ &= \int_{-1}^{1} \lambda_n P_n(t) P_m(t) \, \mathrm{d}t. \end{aligned}$$

We can plug in our previous equation to obtain

$$\lambda_n \langle P_n, P_m \rangle = -\int_{-1}^{1} \frac{\mathrm{d}}{\mathrm{d}t} \left(\left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} \right) P_m(t) \, \mathrm{d}t.$$

By the method of integration by parts, we have

$$\begin{aligned} -\int_{-1}^{1} \frac{\mathrm{d}}{\mathrm{d}t} \left(\left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} \right) P_m(t) \, \mathrm{d}t &= -\left(\left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} P_m(t) \bigg|_{-1}^{1} - \int_{-1}^{1} \left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} \frac{\mathrm{d}P_m}{\mathrm{d}t} \, \mathrm{d}t \right) \\ &= -\left((1 - 1) \left(\frac{\mathrm{d}P_n}{\mathrm{d}t} P_m \right) \bigg|_{-1}^{1} \right) + \int_{-1}^{1} \left(1 - t^2\right) \frac{\mathrm{d}P_n}{\mathrm{d}t} \frac{\mathrm{d}P_m}{\mathrm{d}t} \, \mathrm{d}t \\ &= \int_{-1}^{1} \left(1 - t^2\right) \frac{\mathrm{d}P_m}{\mathrm{d}t} \frac{\mathrm{d}P_n}{\mathrm{d}t} \, \mathrm{d}t. \end{aligned}$$

Using our previous equation, it follows that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\left(1-t^2\right)\frac{\mathrm{d}P_m}{\mathrm{d}t}\right) = -\lambda_m P_m(t).$$

By the method of integration by parts, it follows that

$$\begin{aligned}
\int_{-1}^{1}\left(1-t^2\right)\frac{\mathrm{d}P_m}{\mathrm{d}t}\frac{\mathrm{d}P_n}{\mathrm{d}t}\,\mathrm{d}t &= \left(\left(1-t^2\right)\frac{\mathrm{d}P_m}{\mathrm{d}t}P_n(t)\right)\Bigg|_{-1}^{1} - \int_{-1}^{1}\frac{\mathrm{d}}{\mathrm{d}t}\left(\left(1-t^2\right)\frac{\mathrm{d}P_m}{\mathrm{d}t}\right)\frac{\mathrm{d}P_n}{\mathrm{d}t}\,\mathrm{d}t \\
&= (1-1)\left(\frac{\mathrm{d}P_m}{\mathrm{d}t}P_n(t)\right)\Bigg|_{-1}^{1} + \int_{-1}^{1}\lambda_m P_m(t)P_n(t)\,\mathrm{d}t \\
&= \lambda_m\int_{-1}^{1}P_m(t)P_n(t)\,\mathrm{d}t \\
&= \lambda_m\left\langle P_m, P_n\right\rangle \\
&= \lambda_m\left\langle P_n, P_m\right\rangle.
\end{aligned}$$

Therefore, the Legendre polynomials are mutually orthogonal.

Supplimental 2

Suppose that

$$f(t) = \sum_{n=0}^{\infty} c_n P_n$$

for some $\{c_n\}_{n=0}^{\infty} \subset \mathbb{R}$, and that this series converges for some $t \in \mathbb{R}$. Fix some arbitrary $k \in \mathbb{N}$. Given that the Legendre polynomials are mutually orthogonal, it follows that

$$\begin{aligned}
\left\langle P_k(t), f(t)\right\rangle &= \left\langle P_k(t), \sum_{n=0}^{\infty} c_n P_n(t)\right\rangle \\
&= \int_{-1}^{1} P_k(t)\sum_{n=0}^{\infty} c_n P_n(t)\,\mathrm{d}t \\
&= \sum_{n=0}^{\infty}\int_{-1}^{1} P_k(t) c_n P_n(t)\,\mathrm{d}t \\
&= c_k\int_{-1}^{1}\left(P_k(t)\right)^2\,\mathrm{d}t \\
&= c_k\left\|P_k(t)\right\|_2^2,
\end{aligned}$$

where $\|\cdot\|_2$ is the standard norm on L^2. It follows that

$$c_k = \frac{\left\langle P_k(t), f(t)\right\rangle}{\left\|P_k(t)\right\|_2^2}.$$

Therefore,

$$f(t) = \sum_{n=0}^{\infty}\frac{\left\langle P_n(t), f(t)\right\rangle}{\left\|P_k(t)\right\|_2^2}P_n(t).$$

Supplimental 3